

Ke Zhou

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Education

Hong Kong University of Science and Technology

Master of Philosophy in Computer Science

Research Interest: VR/AR/XR, Human-Computer Interaction, Computer graphics

Sep 2025 – Present

Guangzhou, China

Chongqing University

Bachelor of Economics in Finance (GPA: 3.71/4, Rank: 5/39)

Sep 2021 – Jun 2025

Chongqing, China

Publication

Published (Conferences and Journal)

Handows: A Palm-Based Interactive Multi-Window Management System in Virtual Reality

[Project Link](#)

Jin-Du Wang, [Ke Zhou](#), Haoyu Ren, Per Ola Kristensson, Xiang Li

Accepted by *IEEE TVCG (ISMAR'25)* **Honorable Mention**

Exploring the Remapping Impact of Spatial Head-hand Relations in Immersive Telesurgery

[Project Link](#)

Tianren Luo, [Ke Zhou](#), Pengxiang Wang, Shuting Chang, Gaozhang Chen, Hechuan Zhang, Qi Wang, Teng Han, Feng Tian

Accepted by *ACM CHI'25*

Continuous Measurement Methods for Transient Physiological Discomfort in VR Locomotion

[Project Link](#)

Tianren Luo, Pengxiang Wang, Shuting Chang, [Ke Zhou](#), Nianlong Li, Yulong Bian, Xiaohui Tan, Qi Wang, Teng Han, Feng Tian

Accepted by *ACM CHI'26*

Under Review

GOATUI: Generating Opportunistic Adaptive Tangible User Interfaces through Function Alignment in Augmented Reality

[Project Link](#)

[KeZhou](#), Jin-Du Wang, Tianren Luo, Linping Yuan, Wai Tong

Under Review

Research Experience

VVAI Lab, Texas A&M University

Research Intern advised by **Prof. Wai Tong**

Oct 2024 – Sep 2025

College Station, Texas/Remote

• **User-Edge-Cloud AR System for VLM-Driven Opportunistic Tangible UI Generation.**

- Conducted a formative study with 8 participants to identify key barriers in utilizing daily objects for interaction, synthesizing three design requirements for automated Tangible UI generation.
- Designed and implemented GOATUI, a vision-based perception AR system integrating edge-deployed Lang-SAM (VLM + GroundingDINO + SAM2) for open-vocabulary segmentation, a depth reprojection algorithm to generate 3D interaction realms for gesture recognition and cloud-based VLMs for object-interaction inference.
- Conducted a comparative user study involving 12 participants against manual and surface-based baselines, demonstrating that GOATUI significantly improved interaction expressiveness and user willingness while reducing mental load.

- **Immersive Telesurgery System for Assessing Remapping Effects in VR Teleoperation.**
 - Investigated visual-proprioceptive conflicts arising from discrete spatial remapping during camera adjustments and robotic arm switching, identifying key sensory conflict variables in telesurgical environments.
 - Simulated a high-fidelity immersive telesurgery system in Unity, modeling 18 specific remapping situations across varying offsets and deflections to isolate their impact on surgical precision.
 - Conducted a user study involving 20 participants measuring trajectory deviation and workload, revealing that head-camera deflection induces the most severe spatial perception bias while hand-arm offset causes the highest physical load.
- **Optimized Discomfort Measurement Methods for VR Locomotion.**
 - Supported a user-elicitation study with 24 participants to generate 216 measurement methods, identifying intuitive gestures for reporting real-time motion sickness without breaking immersion.
 - Contributed to the system implementation by developing the VR locomotion scenarios in Unity and integrating the motion capture pipeline using Qualysis for gesture recognition.
 - Facilitated a user study involving 18 participants demonstrating that the proposed method achieved high reliability and minimal cognitive load, providing a validated tool for VR comfort assessment.

- **Immersive Windows Management System in VR/AR.**
 - Identified the physical fatigue and lack of haptics in existing VR window management, and proposed a palm-based interface strategy leveraging proprioception and mobile-inspired gestures.
 - Designed and implemented the Handows system in Unity with Meta Interaction SDK, optimizing spatial layout based on ergonomic angular comfort zones to support selection, closure, positioning, and scaling tasks while providing haptic feedback.
 - Conducted a controlled user study with 15 participants and a case study with 8 participants, results showed Handows reduced head rotation by over 60% and achieved superior scaling precision (1.39% deviation) compared to controller-based baselines.

Awards and Honors

HKUSTGZ Postgraduate Scholarship	2025-2027
Chongqing University Comprehensive Excellence Award	2022-2024
Second Prize in The Chinese Mathematics Competition	2023
Third Prize in National English Competition for College Students	2022
Chongqing University Outstanding Student	2022

Academic Activities

Oral Presentation at CHI 2025	Yokohama, Japan
Oral Presentation at ISMAR 2025	Daejeon, South Korea

Skills

Technical Proficiency: Python, Java, C++, C#, Javascript, HTML, CSS, L^AT_EX, Unity3D (XR Dev), PyTorch
HCI Research Design: User Study Design, Prototyping, Qualitative & Quantitative Data Analysis
Language: Chinese (Native Proficiency), English (Professional Working Proficiency, TOEFL: 104 (R:29, L:27, S:23, W:25))